

Prompt Confusion Phase 04 Arbitration Break Report

Checkpoint memo for the first full-sequence section-attribution and causal-patching pass on April 10, 2026.

MAIN READ

The full-sequence arbitration pass found a real conflict-side signal inside the prompt, with the cleanest policy-source readout in the SETTINGS section at layer 24. The first causal generation pass did not validate yet because the harness returned reasoning-style <think> completions instead of parseable JSON actions.

Run Summary

Capture run

7e71cf742002

123 conflict-resolution rows with full-sequence residual capture.

Section probe

123 rows

Grouped by arbitration_group

Best policy readout

SETTINGS mean @ L24

0.7647 balanced accuracy.

Causal sweep

3 conditions

Baseline plus signed section-direction patches.

This checkpoint uses the conflict-only publication workflow_dataset_conflict_probe_v3_conflict_readout_side_v1 and the downloaded analysis outputs in projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/conflict_arbitration_stage1.

Section Attribution

The strongest policy-source readouts from the residual stream were:

Section	Pooling	Layer	Balanced Accuracy	Interpretation
SETTINGS	mean	24	0.7647	Best policy-source readout selected for direction construction.

SETTINGS	eos	36	0.7639	Nearly tied prompt-end section anchor.
STRATEGY	mean	36	0.7426	Clear strategy-side readout, but weaker than settings.
TASK	eos	36	0.7157	Some arbitration spillover into the instruction boundary.

Interpretation

The strongest overall residual section readouts were actually PORTFOLIO mean @ L28 = 0.7924 and MARKET mean @ L24 = 0.7844, which is a useful confound warning. For the arbitration-specific direction, the right choice is still the best policy-source section rather than the absolute best section, because the goal is to separate SETTINGS versus STRATEGY, not merely to decode contextual pressure.

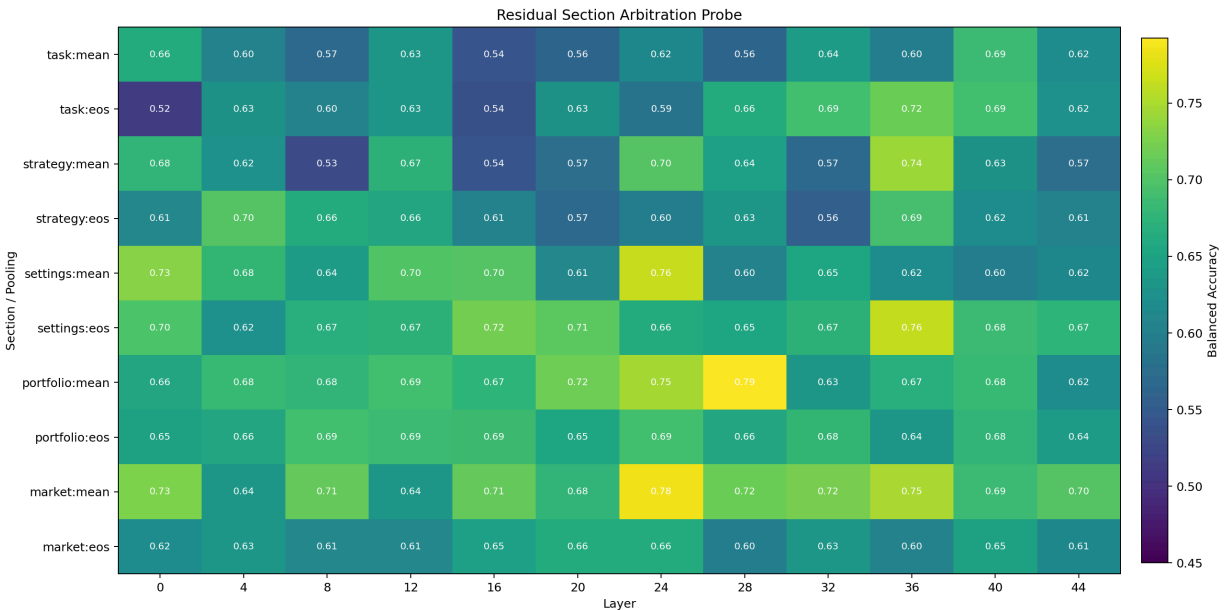


Figure 1: Residual grouped balanced accuracy by section, pooling target, and captured layer.

Router Note

Router section attribution did not produce a usable sweep in this pass. Only 25 rows had router tensors whose token coverage still aligned cleanly with the rendered full prompt spans. That is enough to say the router path is currently partial coverage for this full-sequence setup, but not enough to support a stable grouped section analysis.

Causal Check

The causal patch run targeted the selected arbitration feature:

- section: SETTINGS

- pooling anchor: mean
- layer: 24
- patch strength: 1.0

The run completed operationally, but the behavioral outputs were not usable. All three conditions returned `neither_rate = 1.0` because the model emitted reasoning-style `<think>` completions instead of the expected JSON action object, so the row-level classifier could not map them onto strategy or setting.

Why this is not a null result

The failed causal readout does not mean the patch direction was ineffective. The row-level artifacts show truncated reasoning traces starting with `<think>` rather than valid JSON, which means the causal harness did not yet reproduce the same non-thinking generation surface used in the earlier behavior audit. This is a generation-harness mismatch, not evidence against the arbitration feature.

Local Artifacts

The current checkpoint is downloaded locally:

- `projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/conflict_arbitration_stage1/section_attribution/summary.json`
- `projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/conflict_arbitration_stage1/section_attribution/section_probe_results.parquet`
- `projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/conflict_arbitration_stage1/section_attribution/residual_section_heatmap.png`
- `projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/conflict_arbitration_stage1/causal_check/summary.json`
- `projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/conflict_arbitration_stage1/causal_check/row_level.parquet`

Next Step

The immediate follow-up is narrow:

1. rerun the causal harness with thinking explicitly disabled so baseline generation returns parseable JSON again
2. keep the same selected arbitration feature unless the clean rerun says otherwise: `SETTINGS mean @ layer 24`
3. only after that clean rerun compare baseline versus signed patch shifts by `readout_side` and by family

This checkpoint is enough to justify the next causal iteration, but not enough to claim a successful behavioral intervention yet.